IASTAM 6.0 · TECHNICAL CHALLENGE

BUILDING THE FUTURE BEYOND EARTH

MIZAN: Measure-First Adaptive Inference for Energy-Constrained Satellite Payloads

TEAM INTERSTELLAR · TRACK 1 · PROBLEM 2: AI INFERENCE UNDER ENERGY CONSTRAINTS

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> *A cost profiled on a laptop is not the cost in orbit.*
> **MIZAN measures the board it flies on, and keeps measuring.**

01 CONTEXT & PROBLEM

- Compute is moving to orbit: onboard AI shares a **small energy, memory and time budget** that shifts every orbit.
- Inference can adapt along **three levers**: which model to run, how often, and how much detail to process.
- **The gap:** costs profiled on a laptop are trusted on the payload. We test which lever pays **on the flight board**.

Model · exit depth 1, 2, 3 | **Frequency · run or skip each frame** | **Quality · 48 or 64 px input**

02 OBJECTIVES & ADDED VALUE

- **Objective:** adapt model, frequency or quality to the resources available (compute, memory, energy, deadlines), **never breaking a safety limit**.
- **Demonstrator:** MIZAN on a **Raspberry Pi 4B**, the computing board of the BUPT-1 CubeSat [6].

Accuracy | **Inference time** | **Energy** | **Memory** | **Robustness**

RELATED WORK & ADDED VALUE

Work	Tested on	Adapts to	Energy evidence
Φ-Sat-1 [1]	Myriad 2, in orbit	nothing (fixed filter)	payload power only
Kodan [2]	Jetson Orin	frame deadline	none (time only)
Early exit [3]	host CPU or GPU	input difficulty	FLOPs or latency
E3C [4]	Raspberry Pi 4B	input confidence	metered, fixed draw included
Equinox [5]	simulated constellation	battery, heat, queue	simulated
MIZAN	**Pi 4B + A40, L4, T4**	**deadline, energy, memory, heat**	**measured, above idle**

Added value: one closed loop that uses costs **measured on the target and corrected online**, a safety filter that **can only veto**, and energy split from the board's fixed draw.

03 PROPOSED APPROACH

1 · Measured menu — 8 operating points, costed on the board › **2 · Priced controller** — quality − λ·energy, λ computed onboard › **3 · Veto-only filter** — can refuse, never chooses › **4 · Cost feedback** — each frame corrects the menu

MAIN SYSTEM / ARCHITECTURE

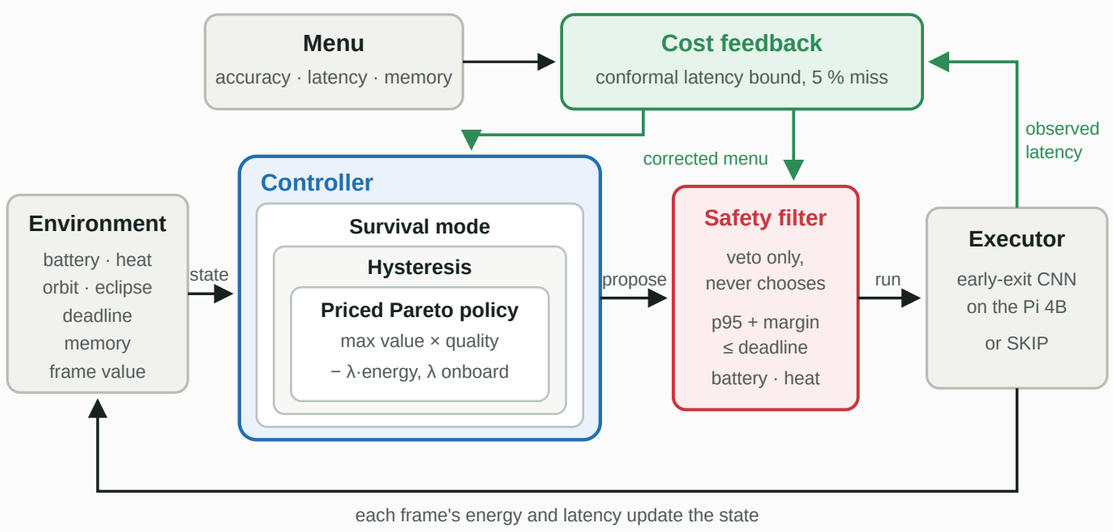
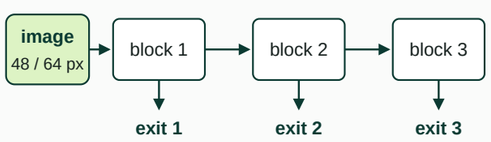


Fig. 1. The MIZAN loop: every executed frame corrects the menu the controller and the filter use.

04 IMPLEMENTATION / PROTOTYPE

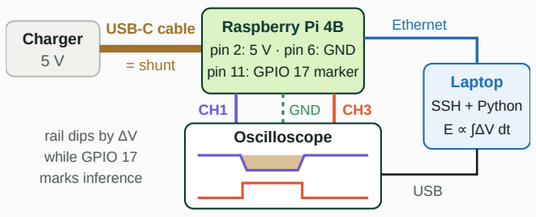
PROTOTYPE / MODEL

confident? stop early, otherwise go deeper



- MobileNetV3-Small [8], two early exits, EuroSAT RGB [7]
- **8 operating points:** 69.3–90.7 % accuracy, 306–315 MB

WORKFLOW / TESTBED



- **Pi 4B** headless, driven over Ethernet
- **Shunt** calibrated by a 0.7 A stress run
- **A40, L4, T4** GPUs; simulator, 7 scenarios

05 METHODOLOGY & EVALUATION

EVALUATION AGAINST THE FIVE INDICATORS

Accuracy	2,000 held-out images	**69.3–90.7 %**
Inference time	p95 on the laptop menu; the Pi runs 6.5× slower	**2.8–9.6 ms**
Energy	GPU counters, saturated, above idle	**52–57 % saved**
Memory	resident memory of the option	**306–315 MB**
Robustness	share of the optimum where deadlines bind	**99.3–99.6 %**

Baselines: every fixed option and an offline optimum. **Protocol:** paced runs, interleaved arms, permutation tests.

Evidence labels: MEASURED · SIMULATED · ESTIMATED · ASCRIBED · INFERRED

06 RESULTS

52–57 % — energy saved per inference, saturated GPUs; ≈ 0 at 10 fps — MEASURED

6.5× — laptop menu too optimistic on the Pi (latency floors) — MEASURED

−49 % — missed deadlines with cost feedback (p = 0.0065, synthetic frames) — MEASURED

0.2–3.4 % — of a paced frame's energy is inference on the Pi — ESTIMATED

Board validation: same decision as the laptop on 100 % of images, all 10 classes; full depth costs ≈ 3× the energy of an early exit (2.6–3.1×, oscilloscope) MEASURED

KEY RESULT: WHICH LEVER PAYS, AND WHERE

deadline slack	duty-cycled payload	saturated accelerator
deadline binds	**Model lever, for deadline safety** — Holds the deadline where the best fixed option fails. SIMULATED MEASURED	**Model lever, on time and energy** — A shorter path cuts latency and energy. MEASURED
deadline slack	**Platform lever, not the model** — Duty cycling and the CPU governor decide energy. INFERRED	**Model lever, for energy** — 52–57 % less energy per inference. MEASURED

ACCURACY UNDER DEADLINE PRESSURE SIMULATED



Fig. 2. MIZAN tracks the upper bound; the most accurate fixed option collapses. Mean of 5 seeds.

07 CONCLUSION & NEXT STEPS

On a duty-cycled payload, adapting the model buys deadline safety, not energy, and only with costs measured on the real board.

- **Win:** where deadlines bind, 99.3–99.6 % of the optimum vs 96.4–96.5 % for the best fixed option SIMULATED
- **Limit:** the cheapest fixed option still misses fewer deadlines on today's narrow menu (0 vs 23) MEASURED
- **Roadmap:** joules on the board · wider M3 menu in the loop · power state as an action · one frozen held-out evaluation

08 REFERENCES / CONTACT

[1] G. Giuffrida *et al.*, "The Φ-Sat-1 mission," *IEEE TGRS*, 2022.
[2] B. Denby *et al.*, "Kodan," *ASPLOS*, 2023.
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[4] Y. Zhao, S. Li, X. Feng, "E3C: early-exit scene classification on edge devices," *ACM MM*, 2025.
[5] A. K. Erol, D. Mahajan, "Equinox," arXiv:2604.19958, 2026.
[6] R. Xing *et al.*, "Satellite computing with COTS devices," *MobiCom*, 2024.
[7] P. Helber *et al.*, "EuroSAT," *IEEE JSTARS*, 2019.
[8] A. Howard *et al.*, "Searching for MobileNetV3," *ICCV*, 2019.

PAPER, DEMO AND POSTER

MIZAN (ميزان) means *balance* in Arabic: accuracy weighed against the energy and time it costs.